The Endocrine System

REIKA ATSUMI, M.D.
MARIA NOVIEL DOMINGUEZ, MD

At the end of the session, students are expected to:

- ❑ Enumerate functions of endocrine system
- ❑ Define hormones
- ❑ Enumerate endocrine glands, their hormones and their functions

ENDOCRINE SYSTEM FUNCTIONS

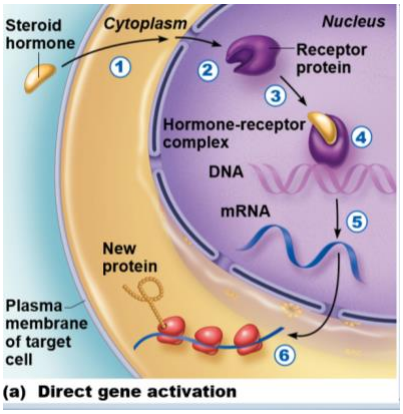
- Second controlling system of the body
  - Nervous system is the fast-control system
- Uses chemical messengers (hormones) that are released into the blood
- Hormones control several major processes
  - Reproduction
  - Growth and development
  - Mobilization of body defenses
  - Maintenance of much of homeostasis
  - Regulation of metabolism

HORMONES

- Hormones are produced by specialized cells
- Cells secrete hormones into extracellular fluids
- Blood transfers hormones to target sites
- These hormones regulate the activity of other cells
- Endocrinology is the scientific study of hormones and endocrine organs

The Chemistry of Hormones

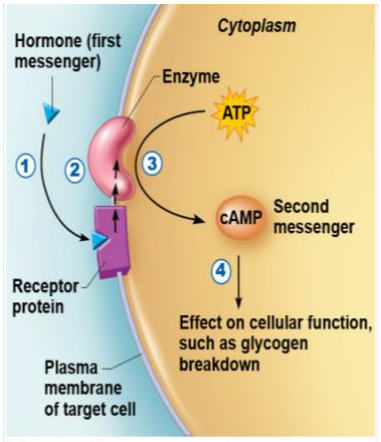
- Hormones are classified chemically as:
  - Amino acid–based, which includes:
    - Proteins
    - Peptides
    - Amines
  - Steroids—made from cholesterol
  - Prostaglandins—made from highly active lipids that act as local hormones



(a) Direct gene activation

Hormone Action

- Hormones affect only certain tissues or organs (target cells or target organs)
- Target cells must have specific protein receptors
- Hormone binding alters cellular activity
- Hormones arouse cells or alter cellular activity
- Typically, one or more of the following occurs:
  - Change plasma membrane permeability or membrane potential by opening or closing ion channels
  - Activate or inactivate enzymes
  - Stimulate or inhibit cell division
  - Promote or inhibit secretion of a product
  - Turn on or turn off transcription of certain genes
- Hormones act by two mechanisms
  - ✓ Direct gene activation
    - Used by steroid hormones and thyroid hormone
      1. Steroid hormones diffuse through the plasma membrane of target cells
      2. Once inside the cell, the hormone enters the nucleus
      3. Then, the hormone binds to a specific protein within the nucleus
      4. Hormone-receptor complex binds to specific sites on the cell's DNA
      5. Certain genes are activated to transcribe messenger RNA
      6. New proteins are synthesized
  - ✓ Second-messenger system
    - Used by protein and peptide hormones
      1. Hormone (first messenger) binds to a membrane receptor
      2. Activated receptor sets off a series of reactions that activates an enzyme
      3. Enzyme catalyzes a reaction that produces a second-messenger molecule (such as cyclic AMP, known as cAMP)



(b) Second-messenger system

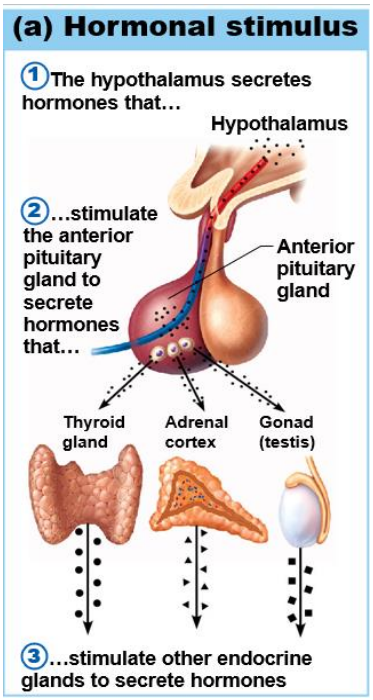
© 2018 Pearson Education, Inc.

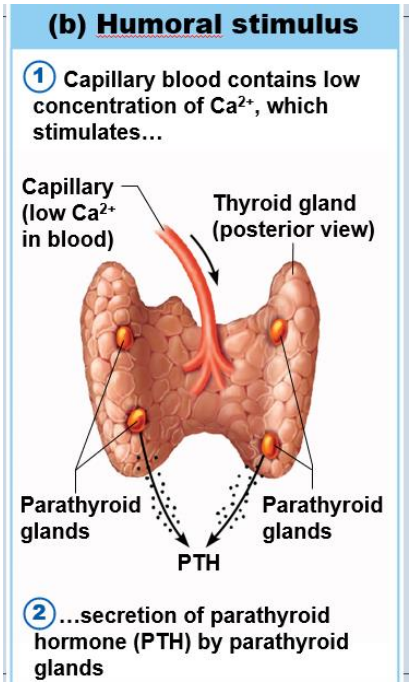
4. Oversees additional intracellular changes to promote a specific response in the target cell

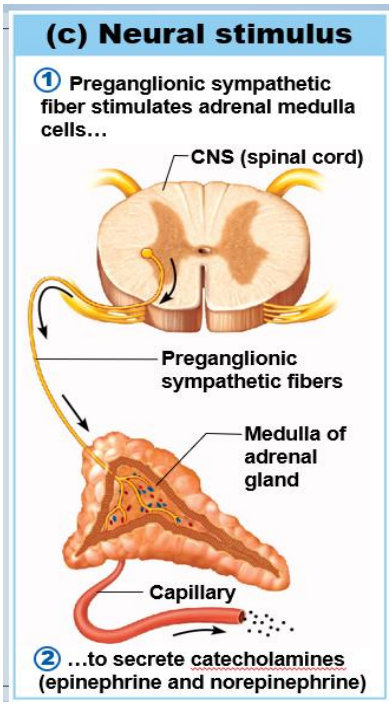
Stimuli for Control of Hormone Release

- Hormone levels in the blood are maintained mostly by negative feedback
  - A stimulus or low hormone levels in the blood trigger the release of more hormone
  - Hormone release stops once an appropriate level in the blood is reached
- The stimuli that activate endocrine glands fall into three major categories

Hormonal	Humoral	Neural
▪ Most common category of stimulus ▪ Endocrine organs are activated by other hormones	▪ Changing blood levels of certain ions and nutrients stimulate hormone release ▪ *Humoral* indicates various body fluids, such as blood and bile	• Nerve fibers stimulate hormone release • Most are under the control of the sympathetic nervous system
Example: ✓ Hormones of the hypothalamus stimulate the anterior pituitary to secrete its hormones	Examples: ✓ Parathyroid hormone and calcitonin are produced in response to changing levels of blood calcium levels ✓ Insulin is produced in response to changing levels of blood glucose levels	Examples: ✓ Sympathetic stimulation of the adrenal medulla to release epinephrine and norepinephrine

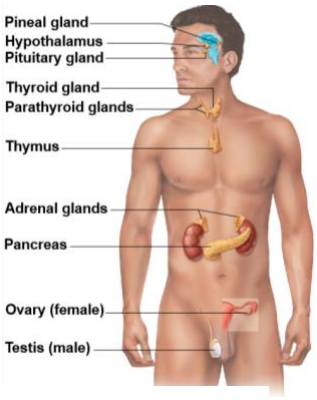






© 2018 Pearson Education, Inc.

ENDOCRINE GLANDS, THEIR HORMONES AND THEIR FUNCTIONS



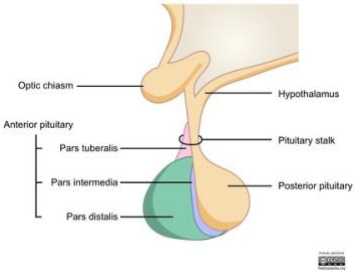
The Major Endocrine Organs
Hypothalamus
Pituitary gland
Pineal gland
Thyroid gland
Parathyroid glands
Thymus
Adrenal glands
Pancreas
Gonads (testes and ovaries)

Some glands have purely endocrine functions , ductless glands
- Anterior pituitary, thyroid, adrenals, parathyroids

Other glands are mixed glands, with both endocrine and exocrine functions (pancreas, gonads)
- Pancreas, gonads

Hypothalamus

- Produces releasing hormones and inhibiting hormones
  - These hormones are released into portal circulation, which connects hypothalamus to anterior pituitary
- Hypothalamus also makes two hormones: oxytocin and antidiuretic hormone
  - Carried to posterior pituitary via neurosecretory cells for storage

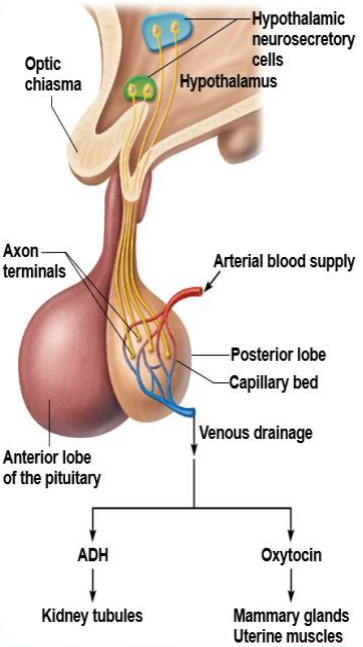


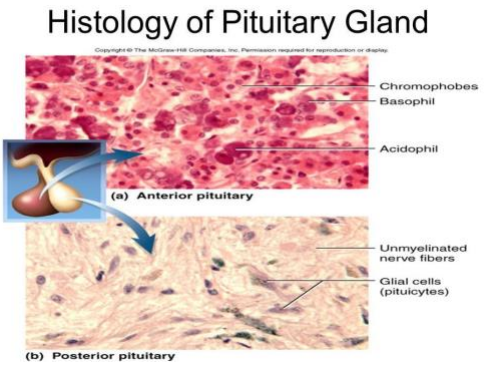
Pituitary Gland

- Pea-sized gland that hangs by a stalk from the hypothalamus in the brain
- Has two functional lobes
  - Anterior pituitary—glandular tissue
  - Posterior pituitary—nervous tissue
- Often called the "master endocrine gland"

✓ Posterior pituitary (Neurohypophysis)
- Does not make the hormones it releases
- Stores hormones made by the hypothalamus
- Two hormones released

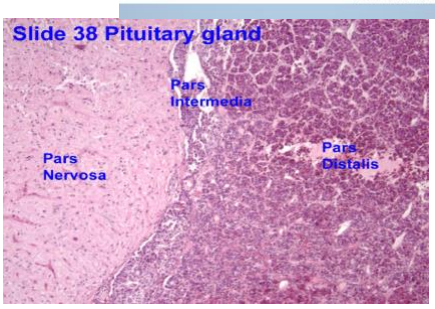
1. Oxytocin
   - ❖ Stimulates contractions of the uterus during labor, sexual relations, and breastfeeding
   - ❖ Causes milk ejection (let-down reflex) in a breastfeeding woman
2. Antidiuretic hormone (ADH)
   - ❖ Inhibits urine production (diuresis) by promoting water reabsorption by the kidneys
   - ❖ Urine volume decreases, blood pressure increases
   - ❖ In large amounts, causes constriction of arterioles, leading to increased blood pressure (the reason why ADH is known as *vasopressin*)
   - ❖ Alcohol inhibits ADH secretion





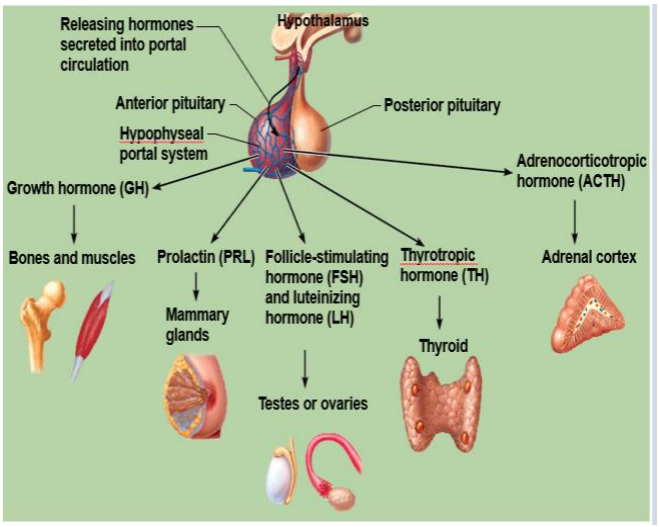
Chromophils

	Acidophils	Basophils
Number	37%	11%
Nature	Protein	Glycoprotein
PAS stain	PAS -ve	PAS +ve
Size	Smaller	Larger
Granules	Larger	Smaller
Secretion	1- Somatotrophs (growth h.) 2-Mammotrophs (prolactin)	1-Thyrotrophs (TSH) 2-Gonadotrophs (FSH& LH) 3-Corticotrophs (ACTH)



© 2018 Pearson Education, Inc.

- ✓ Anterior Pituitary (Adenohypophysis)
  - All anterior pituitary hormones:
  - Are proteins (or peptides)
  - Act through second-messenger systems
  - Are regulated by hormonal stimuli
  - Are regulated mostly by negative feedback
  - Six anterior pituitary hormones
    - ➢ Two hormones affect nonendocrine targets
      1. Growth hormone (somatotrophs)
      2. Prolactin (mammotrophs)
    - ➢ Four are tropic hormones
      1. Follicle-stimulating hormone (gonadotrophs)
      2. Luteinizing hormone (gonadotrophs)
      3. Thyrotropic hormone
      4. Adrenocorticotropic hormone



Growth hormone (GH)
- General metabolic hormone
- Major effects are directed to growth of skeletal muscles and long bones
- Plays a role in determining final body size
- Causes amino acids to be built into proteins
- Causes fats to be broken down for a source of energy

Prolactin (PRL)
- Stimulates and maintains milk production following childbirth
- Function in males is unknown

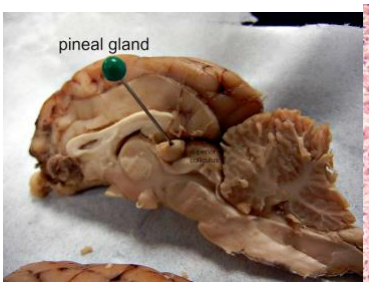
Follicle-stimulating hormone (FSH)
- Stimulates follicle development in ovaries
- Stimulates sperm development in testes

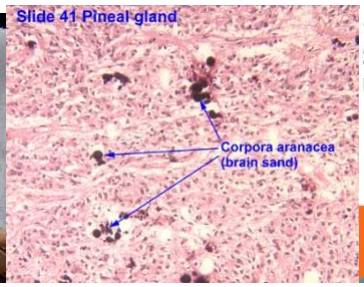
Luteinizing hormone (LH)
- Triggers ovulation of an egg in females
- Stimulates testosterone production in males

Thyrotropic hormone (TH)/ *Thyroid-stimulating hormone (TSH)*
- Influences growth and activity of the thyroid gland

Adrenocorticotropic hormone (ACTH)
- Regulates endocrine activity of the adrenal cortex

Pineal Gland
- Hangs from the roof of the third ventricle of the brain
- Secretes melatonin
  - Believed to trigger the body's sleep/wake cycle
  - Believed to coordinate the hormones of fertility in humans and to inhibit the reproductive system until maturity occurs





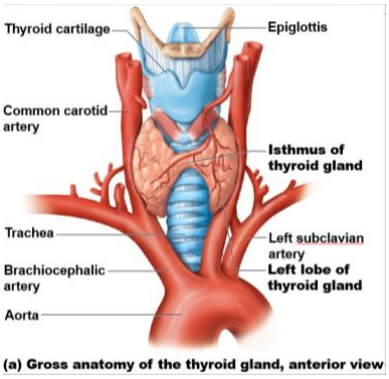


© 2018 Pearson Education, Inc.

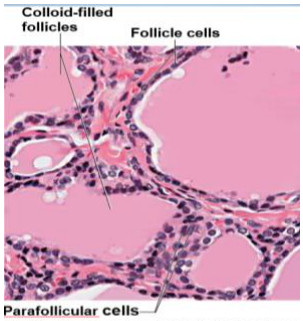
Thyroid Gland

- Found at the base of the throat, inferior to the Adam's apple
- Consists of two lobes and a connecting isthmus
- Follicles are hollow structures that store colloidal material
- Produces two hormones

1. Thyroid hormone
   - Major metabolic hormone
   - Controls rate of oxidation of glucose to supply body heat and chemical energy
   - Needed for tissue growth and development
   - Composed of two active iodine-containing hormones
     - Thyroxine (T_4)—secreted by thyroid follicles
     - Triiodothyronine (T_3)—conversion of T_4 at target tissues
2. Calcitonin
   - Decreases blood calcium levels by causing calcium deposition on bone
   - Antagonistic to parathyroid hormone
   - Produced by parafollicular cells found between the follicles



(a) Gross anatomy of the thyroid gland, anterior view



(b) Photomicrograph of thyroid gland follicles (380x)

Parathyroid Glands

- Tiny masses on the posterior of the thyroid
- Secrete parathyroid hormone (PTH)
  - Most important regulator of calcium ion (Ca^{2+}) homeostasis of the blood
  - Stimulates osteoclasts to remove calcium from bone
  - Hypercalcemic hormone (increases blood calcium levels)
  - Stimulates the kidneys and intestine to absorb more calcium

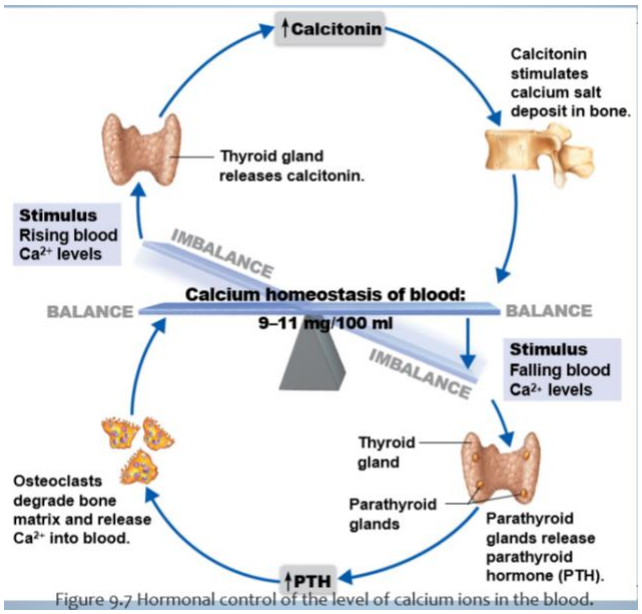


Figure 9.7 Hormonal control of the level of calcium ions in the blood.

Chief cells

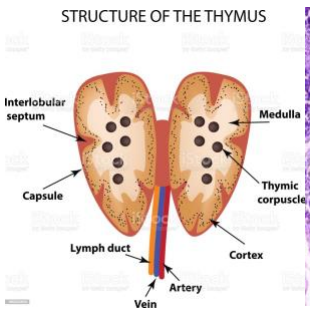
- primary producer of hormone
- Smaller

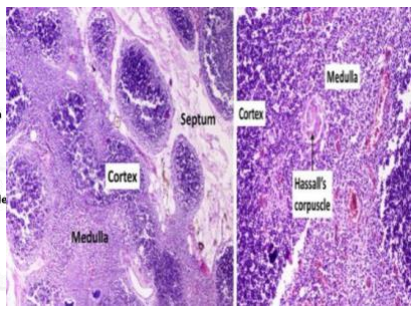
Oxyphil cells

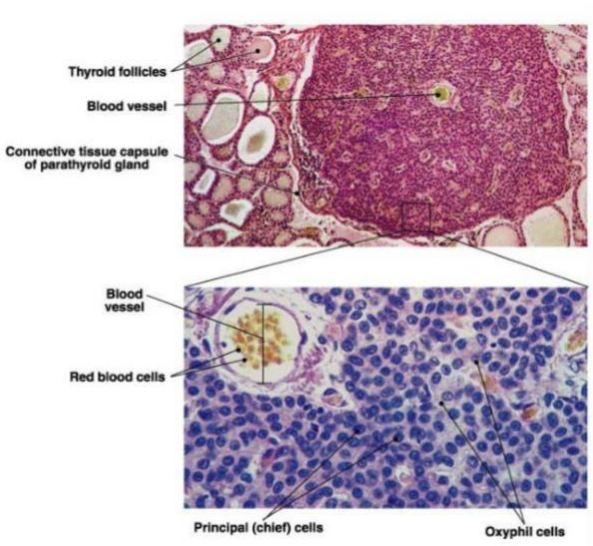
- secondary producers
- contains more cytoplasm
- appears lighter in color

Thymus

- Located in the upper thorax, posterior to the sternum
- Largest in infants and children
- Decreases in size throughout adulthood
- Produces a hormone called *thymosin*
  - Matures some types of white blood cells
  - Important in developing the immune system





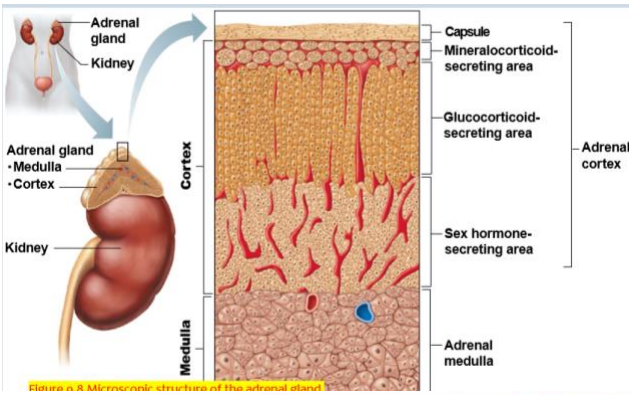


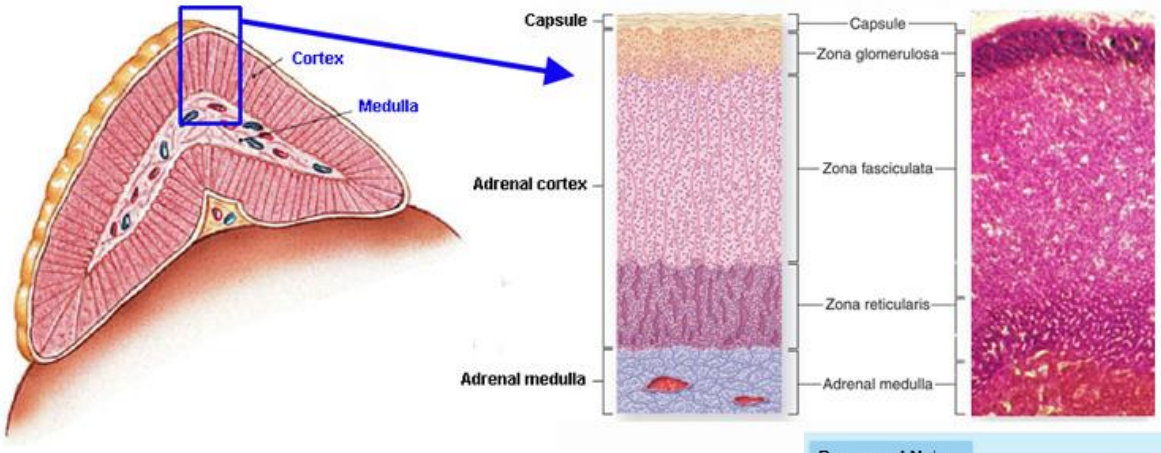
Adrenal Glands

© 2018 Pearson Education, Inc.

- Sit on top of the kidneys
- Two regions

1. Adrenal cortex—outer glandular region has three layers that produce corticosteroids
   a. Mineralocorticoids are secreted by outermost layer
   b. Glucocorticoids are secreted by middle layer
   c. Sex hormones are secreted by innermost layer
2. Adrenal medulla—inner neural tissue region

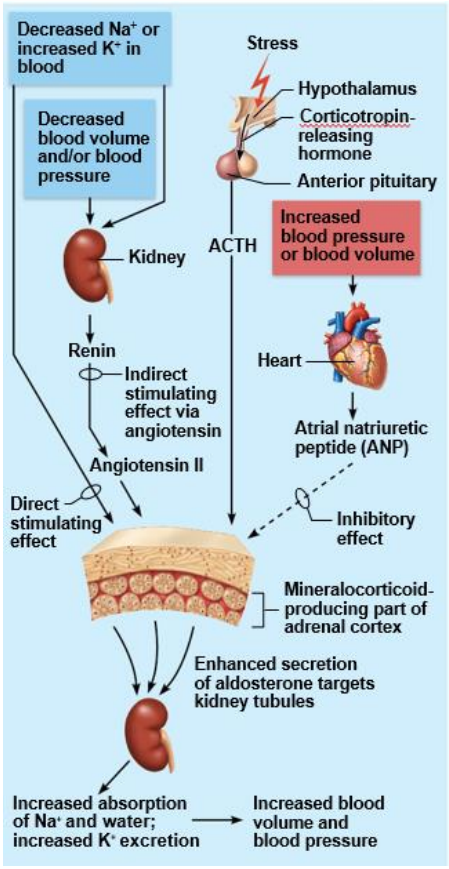




Hormones of the adrenal cortex

1. Mineralocorticoids (mainly aldosterone)
   - Produced in outer adrenal cortex
   - Regulate mineral (salt) content in blood, particularly sodium and potassium ions
   - Regulate water and electrolyte balance
   - Target organ is the kidney
   - Release of aldosterone is stimulated by:
     - Humoral factors (fewer sodium ions or too many potassium ions in the blood)
     - Hormonal stimulation (ACTH)
     - Renin and angiotensin II in response to a drop in blood pressure
   - Aldosterone production is inhibited by atrial natriuretic peptide (ANP), a hormone produced by the heart when blood pressure is too high

2. Glucocorticoids (including cortisone and cortisol)
   - Produced by middle layer of adrenal cortex
   - Promote normal cell metabolism
   - Help resist long-term stressors by increasing blood glucose levels (hyperglycemic hormone)
   - Anti-inflammatory properties
   - Released in response to increased blood levels of ACTH



© 2018 Pearson Education, Inc.

3. Sex hormones
    - Produced in the inner layer of the adrenal cortex
    - Small amounts are made throughout life
    - Most of the hormones produced are androgens (male sex hormones), but some estrogens (female sex hormones) are also formed

Adrenal medulla

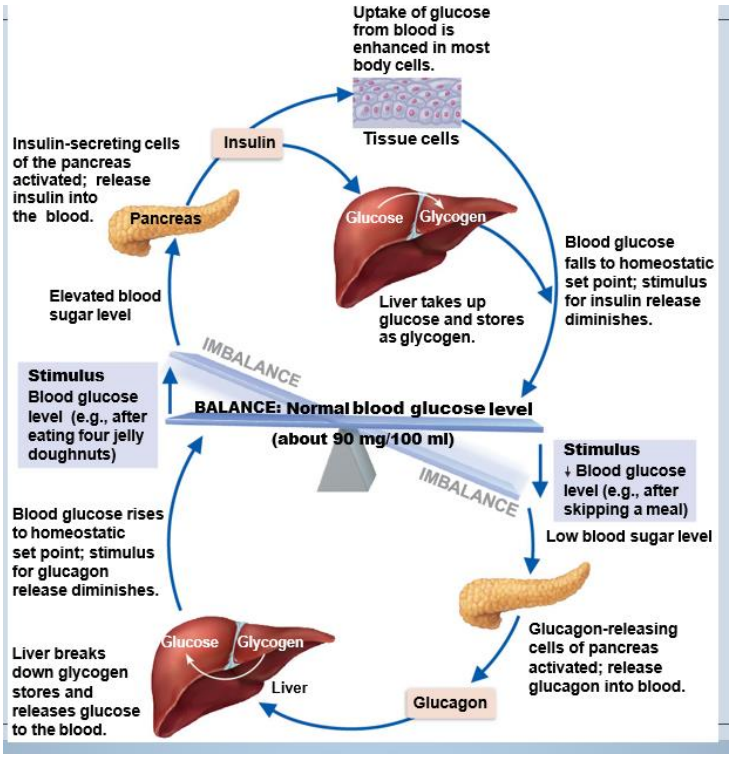
- Produces two similar hormones (catecholamines)

1. Epinephrine (adrenaline)
2. Norepinephrine (noradrenaline)

- These hormones prepare the body to deal with short-term stress ("fight or flight") by:
    - Increasing heart rate, blood pressure, blood glucose levels
    - Dilating small passageways of lungs

Pancreas

- Located in the abdomen, close to stomach
- Mixed gland, with both endocrine and exocrine functions
- The pancreatic islets (islets of Langerhans) produce hormones that are antagonists to maintain blood sugar homeostasis
    1. Insulin—produced by beta cells
        - Released when blood glucose levels are high
        - Increases the rate of glucose uptake and metabolism by body cells
        - Effects are hypoglycemic
    2. Glucagon—produced by alpha cells
        - Released when blood glucose levels are low
        - Stimulates the liver to release glucose to blood, thus increasing blood glucose levels



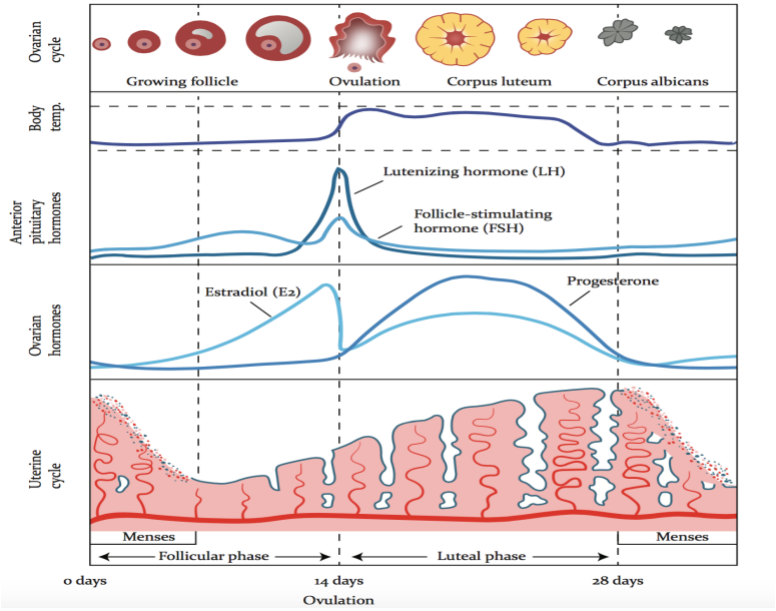
Gonads

- Produce sex cells
- Produce sex hormones

Ovaries

- located in the pelvic cavity, produce eggs
- Produce two groups of steroid hormones

1. Estrogens
    - Stimulate the development of secondary female characteristics
    - Mature the female reproductive organs
    - With progesterone, estrogens also:
        - Promote breast development
        - Regulate menstrual cycle
2. Progesterone
    - Acts with estrogen to bring about the menstrual cycle
    - Helps in the implantation of an embryo in the uterus
    - Helps prepare breasts for lactation



© 2018 Pearson Education, Inc.

Testes

- Male gonads suspended outside the pelvic cavity
- Produce sperm
- Produce androgens, such as testosterone
  - Testosterone
    - the most important androgen
    - Responsible for adult male secondary sex characteristics
    - Promotes growth and maturation of male reproductive system
    - Required for sperm cell production

Secondary sexual characteristics at puberty

Male	Female
Voice becomes much lower (breaks)	Breasts grow, nipples enlarge
Hair start to grow on chest, face, under arms and in pubic area	Hair develops under arms and in pubic area
Body becomes more muscular	Hip become wider
Penis becomes larger	Uterus and vagina become larger
Testers start to produce sperm	Ovaries start to release eggs and period begin (menstruation)

Table 9.1 Major Endocrine Glands and Some of Their Hormones

Gland	Hormone	Chemical class*	Major actions	Regulated by
Hypothalamus 	Hormones released by the posterior pituitary; releasing and inhibiting hormones that regulate the anterior pituitary (see below)			
Pituitary gland				
• Posterior lobe (releases hormones made by the hypothalamus) 	Oxytocin	Peptide	Stimulates contraction of uterus and the milk "let-down" reflex	Nervous system (hypothalamus), in response to uterine stretching and/or suckling of a baby
	Antidiuretic hormone (ADH)	Peptide	Promotes retention of water by kidneys	Hypothalamus, in response to water/salt imbalance
• Anterior lobe 	Growth hormone (GH)	Protein	Stimulates growth (especially of bones and muscles) and metabolism	Hypothalamic releasing and inhibiting hormones
	Prolactin (PRL)	Protein	Stimulates milk production	Hypothalamic hormones
	Follicle-stimulating hormone (FSH)	Protein	Stimulates production of ova and sperm	Hypothalamic hormones
	Luteinizing hormone (LH)	Protein	Stimulates ovaries and testes	Hypothalamic hormones
	Thyroid-stimulating hormone (TSH)	Protein	Stimulates thyroid gland	Thyroxine in blood; hypothalamic hormones
	Adrenocorticotropic hormone (ACTH)	Protein	Stimulates adrenal cortex to secrete glucocorticoids	Glucocorticoids; hypothalamic hormones

* Any class not indicated as steroid is amino acid based.

© 2018 Pearson Education, Inc.

Table 9.1 **Major Endocrine Glands and Some of Their Hormones** (*continued*)

Gland		Hormone	Chemical class*	Major actions	Regulated by
Pineal gland		Melatonin	Amine	Involved in biological rhythms (daily and seasonal)	Light/dark cycles
Thyroid gland		Thyroxine (T_4) and triiodothyronine (T_3)	Amine	Stimulates metabolism	TSH
		Calcitonin	Peptide	Reduces blood calcium ion level	Calcium ion level in blood
Parathyroid glands		Parathyroid hormone (PTH)	Peptide	Raises blood calcium ion level	Calcium ion level in blood

* Any class not indicated as steroid is amino acid based.

Table 9.1 **Major Endocrine Glands and Some of Their Hormones** (*continued*)

Gland		Hormone	Chemical class*	Major actions	Regulated by
Thymus		Thymosin	Peptide	"Programs" T lymphocytes	Not known
Adrenal glands					
• Adrenal medulla		Epinephrine and norepinephrine	Amines	Raise blood glucose level; increase rate of metabolism; constrict certain blood vessels	Nervous system (sympathetic division)
• Adrenal cortex		Glucocorticoids	Steroids	Increase blood glucose level	ACTH
		Mineralocorticoids	Steroids	Promote reabsorption of Na^+ and excretion of K^+ (potassium) in kidneys	Changes in blood volume or blood pressure; K^+ or Na^+ level in blood
		Androgens and estrogens (see entry under gonads)			

* Any class not indicated as steroid is amino acid based.

© 2018 Pearson Education, Inc.

Table 9.1 **Major Endocrine Glands and Some of Their Hormones** (*continued*)

Gland	Hormone	Chemical class*	Major actions	Regulated by
Pancreas 	Insulin	Protein	Reduces blood glucose level	Glucose level in blood
	Glucagon	Protein	Raises blood glucose level	Glucose level in blood
Gonads				
• Testes 	Androgens	Steroids	Support sperm formation; development and maintenance of male secondary sex characteristics	FSH and LH
• Ovaries 	Estrogens	Steroids	Stimulate uterine lining growth; development and maintenance of female secondary sex characteristics	FSH and LH
	Progesterone	Steroids	Promotes growth of uterine lining	FSH and LH

* Any class not indicated as steroid is amino acid based.

Other Hormone-Producing Tissues and Organs

Other organs that are generally nonendocrine in function also secrete hormones

- ✓ Stomach
- ✓ Small intestine
- ✓ Kidneys
- ✓ Heart
- ✓ Placenta
  - Produces hormones that maintain pregnancy
  - Some hormones play a part in the delivery of the baby
    - Produces human chorionic gonadotropin (hCG) in addition to estrogen, progesterone, and other hormones
    - Human placental lactogen (hPL) prepares the breasts for lactation
    - Relaxin relaxes pelvic ligaments and pubic symphysis for childbirth

Table 9.2 **Hormones Produced by Organs Other Than the Major Endocrine Organs**

Hormone	Chemical composition	Source	Stimulus for secretion	Target organ/Effects
Prostaglandins (PGs); several groups indicated by letters A–I (PGA–PGI)	Derived from fatty acid molecules	Plasma membranes of virtually all body cells	Various (local irritation, hormones, etc.)	Have many targets but act locally at site of release. Examples of effects include the following: increase blood pressure by acting as vasoconstrictors; cause constriction of respiratory passageways; stimulate muscle of the uterus, promoting menstrual pain and labor; enhance blood clotting; promote inflammation and pain; increase output of digestive secretions by stomach; cause fever.
Gastrin	Peptide	Stomach	Food	*Stomach:* stimulates glands to release hydrochloric acid (HCl).

© 2018 Pearson Education, Inc.

Table 9.2 **Hormones Produced by Organs Other Than the Major Endocrine Organs** (*continued*)

Hormone	Chemical composition	Source	Stimulus for secretion	Target organ/Effects
Intestinal gastrin	Peptide	Duodenum (first segment) of small intestine	Food, especially fats	*Stomach:* stimulates gastric glands and motility.
Secretin	Peptide	Duodenum	Food	*Pancreas:* stimulates release of bicarbonate-rich juice. *Liver:* increases release of bile. *Stomach:* reduces secretions and motility.
Cholecystokinin (CCK)	Peptide	Duodenum	Food	*Pancreas:* stimulates release of enzyme-rich juice. *Gallbladder:* stimulates expulsion of stored bile. *Duodenal papilla* (entry point at the small intestine): causes sphincter to relax, allowing bile and pancreatic juice to enter duodenum.
Erythropoietin	Glycoprotein	Kidney	Hypoxia (low	*Bone marrow:* stimulates production of

Table 9.2 **Hormones Produced by Organs Other Than the Major Endocrine Organs** (*continued*)

Hormone	Chemical composition	Source	Stimulus for secretion	Target organ/Effects
Active vitamin D_3	Steroid	Kidney (activates provitamin D made by epidermal cells)	PTH	*Intestine:* stimulates active transport of dietary calcium ions across intestinal cell membranes.
Atrial natriuretic peptide (ANP)	Peptide	Heart	Stretching of atria (receiving chambers) of heart	*Kidney:* inhibits sodium ion reabsorption and renin release. *Adrenal cortex:* inhibits secretion of aldosterone, thereby decreasing blood volume and blood pressure.
Leptin	Peptide	Adipose tissue	Fatty foods	*Brain:* suppresses appetite and increases energy expenditure.
Resistin	Peptide	Adipose tissue	Unknown	*Fat, muscle, liver:* antagonizes insulin's action on liver cells.

Developmental Aspects of the Endocrine System

- In the absence of disease, efficiency of the endocrine system remains high until old age
- Decreasing function of female ovaries at menopause leads to such symptoms as osteoporosis, increased chance of heart disease, and possible mood changes
- Developmental Aspects of the Endocrine System
- Efficiency of all endocrine glands gradually decreases with aging, which leads to a generalized increase in incidence of:
  - Diabetes mellitus
  - Immune system depression
  - Lower metabolic rate
  - Cancer rates in some areas

© 2018 Pearson Education, Inc.